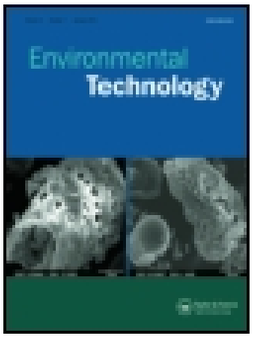

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Enhanced fluoride removal behavior and mechanism by dicalcium phosphate from aqueous solution

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Abstract:

Dicalcium phosphate was prepared by ethylenediaminetetraacetic acid as a calcium chelating agent, and further explored to remove the fluoride ions from aqueous solution. The as-prepared samples main existed in the monetite phase from the result of XRD. The dried sample consisted of small nanoparticles and displayed irregular particles with a size of ca. 3 μm due to the agglomeration. The fluoride removal ability was evaluated by batch adsorption experiments. The as-prepared adsorbent exhibited the enhanced fluoride removal behavior with the maximum adsorption capacity of 66.72 mg/g from the Langmuir isotherm model, which was higher than that of other previously reported calcium phosphate. The adsorbent could be utilized

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in the wide pH range of 3-10. The adsorption kinetics could be better described by the pseudo-second-order model than first-second-order model. The co-existing anions had a negligible influence on the fluoride adsorption. The investigation of adsorption mechanism suggested that the chemical reaction and/or dissolution – precipitation mechanism should be dominant in the fluoride adsorption process, accompanying with electronic interaction and ions exchange, which enhanced the fluoride removal performance.
Keywords: dicalcium phosphate, monetite, fluoride, kinetics, adsorption mechanism.

1. Introduction

The chronic ingestion of excess fluoride in drinking water has harmful influence on human health, such as dental or skeletal fluorosis, brain damage and thyroid disorder [1, 2, 3]. The guideline value of fluoride recommended by World Health Organization (WHO) is 1.5 mg/L, and acceptable concentration is in the range of 0.5-1.5 mg/L, which provides the positive protection against tooth decay [4]. An excess or absence of fluoride in drinking water is detrimental to mankind. Fluoride contamination seems ubiquitous in developing country, such as India, Mexico, Argentina, China and Bengal [4]. The purification of fluoride-contaminated water becomes a global environmental concern in the past decades.

A multitude of defluoridation methods are proposed to eliminate the excess fluoride from water, such as precipitation [5], electrodialysis [6, 7], ion exchange [8], reverse osmosis [9] and adsorption [10, 11, 12, 13]. Most methods suffer from relatively high operation costs, risk of secondary pollutants and complex treatment processes [14]. Adsorption is regarded as the most efficient and applicable approach in defluoridation because it can solve the aforementioned problems [15, 16]. Activated aluminum oxide is often utilized in practical defluoridation application due to the low cost, but its drawback is also distinct, such as relatively low adsorption capacity and weak stability [17, 18]. To date, numerous efforts still have been devoted to explore and engineer the novel adsorbent for defluoridation.

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Calcium phosphate-based materials, which can be essentially divided into cements or ceramics, including calcite, apatite, dicalcium phosphate, hydroxyapatite (HA), have wide applications in bone repair, drug delivery, orthopedics, cancer therapy and biological matrixes due to their non-toxicity and low cost [19]. Recently, there are a large number of attempts to abate the fluoride contamination by calcium phosphate [14, 20, 21, 22]. To some extent it exhibits the outstanding potential as a candidate for fluoride removal. Zhang et al. prepared hydroxyapatite nanoparticles via microwave irradiation technology and found that the fluoride adsorption capacity could reach 19.7 mg/g from Langmuir–Freundlich model [20]. Jayarathne et al. investigated natural apatite with a defluoridation capacity of 0.212 mg/g [22]. Mourabet et al. synthesized the brushite adsorbent to remove the fluoride, and the adsorbent had a maximum monolayer adsorption capacity of 6.59 mg/g [21]. Kanno et al. fabricated a novel nano to micrometer sized HA on the surfaces of granular limestone to improve the adsorption efficiency of the HA-based filter with a fluoride adsorption capacity of 8.8 mg/g, and the mode of adsorption is through the formation of fluoridated-HA or fluorapatite at low fluoride levels and fluorite at high fluoride loadings [14]. Dicalcium phosphate (DCP) is an important class of calcium phosphate, which consists of brushite (DCPD) and monetite (DCPA). Brushite exhibited the weak mechanical property, and easily was transformed to monetite [23], while monetite is stable under acidic conditions (pH below 4.8) [24]. Thus, it is speculated that monetite will be more suitable for use as fluoride adsorbent. Shen et al. fabricated the well-dispersed monetite bundles inlaid in chitosan beads (MONs@CS) with a maximum fluoride adsorption capacity of 50.01 mg/g, and further proposed that the possible mechanism for fluoride removal should be the ion exchange and electrostatic interactions that originated from the pre-protonated amine groups of chitosan matrix [25].

Herein, we reported the fabrication of dicalcium phosphate (main monetite phase) adsorbent, and investigated the defluoridation performance. The adsorbent was

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evaluated in terms of adsorption isotherm, effect of pH, adsorption kinetic. The possible adsorption mechanism was proposed by means of proper characterization.

2. Experimental Section

2.1 Materials

The chemicals are of analytical reagent grade and used without any further purification. Calcium chloride ($CaCl_2$), Ethylenediaminetetraacetic acid (EDTA), Ammonium hydrogen phosphate ($(NH_4)_2HPO_4$), urea were purchased from local chemical agent suppliers. Ultrapure water (18.2 MΩcm) was used for all the experiments.

2.2 Synthesis of dicalcium phosphate (DCP) adsorbent

Dicalcium phosphate adsorbent was synthesized by hydrothermal method using EDTA as a calcium chelating agent. Typically, a solution containing 1 mol/L $CaCl_2$ and 0.1 mol/L EDTA was prepared by dissolving $CaCl_2$ and EDTA in distilled water. Then, 60 mL of above solution and 60 mL of 0.6 mol/L $(NH_4)_2HPO_4$ solution were sequentially added into 24 mL of 1 mol/L urea solution under vigorous stirring. After adjusting the pH value of the mixed solution to pH 4 with 0.1 mol/L HCl, the solution was continuously stirred for 30 min. Subsequently, the solution was transferred into a 200 mL autoclave with a Teflon liner at 180 ºC for 1 h. The resulting precipitate was collected by centrifugation, washed three times with distilled water and dried in vacuum at 80 ºC for 10 h. The samples were denoted as DCP-X, where X means the reaction time.

2.2 Characterization of dicalcium phosphate adsorbent.

The morphology of the adsorbent was observed by scanning electron microscope (SEM, Hitachi SU8020) and transmission electron microscope (TEM, Hitachi 7700). The Brunauer–Emmett–Teller (BET) specific surface of the products was calculated from N_2 adsorption–desorption isotherms at 77 K by an automated gas sorption analyzer (Quantachrome Instruments, autosorb-iQ). The phase characterization of the powders was investigated by X-ray diffraction (XRD) using a Rigaku D/max 2500 diffractometer at a voltage of 40 kV and a current of 200 mA with Cu Kα radiation

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(λ=1.5406 Å), employing a scanning rate 4° min^{-1} in the 2θ ranging from 10 to 65°. X-ray photoelectron spectroscopy (XPS) was performed on the Thermo Scientific ESCALab 250Xi using 200 W monochromatic Al Kα radiation. Typically the hydrocarbon C1s line at 284.8 eV from adventitious carbon is used for energy referencing. The Fourier transform infrared (FTIR) spectra of the obtained samples were recorded with a Tensor 27 FTIR spectrometer in the range of 4000–400 cm^{-1}.

2.3 Batch adsorption experiments

Fluoride mother solution (2000 mg/L) was prepared by dissolving NaF into the distilled water, and the other fluoride solution was diluted from the mother solution. The adsorption experiments were carried out at 200 rpm and 25 ±1 ºC, and the adsorption dose was kept at 1 g/L for all the adsorption experiments. Expect the effect of pH, the initial pH value was not adjusted. The initial fluoride concentration for adsorption kinetic and pH effect experiments was 50 mg/L, while the effect of coexisting ions and effect of hydrothermal time used 36 and 75 mg/L as the initial fluoride concentration, respectively. The adsorption isotherm was performed in the concentration range of 0-200 mg/L. The residual fluoride concentration in the solution was measured by an ion chromatography (ICS-900, Dionex). The adsorption capacity at equilibrium for fluoride was calculated according to the following equation:

$$q_e = \frac{(C_0 - C_e)V}{m} \tag{1}$$

where C_0 (mg/L) and C_e (mg/L) present the initial and equilibrium fluoride concentration, respectively, V (L) is the volume of the fluoride solution and m (g) is the amount of the adsorbent.

The Langmuir isotherm assumes the monolayer adsorption with the uniform energy on the surface of adsorbent, while the Freundlich isotherm described heterogeneous surfaces with nonenergetically equivalent sites, which can be expressed as [15, 26]:

$$q_e = \frac{Q_{\max} b C_e}{1 + b C_e} \tag{2}$$

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$$q_e = K_f C_e^{1/n} \tag{3}$$

where C_e (mg/L) is the equilibrium fluoride concentration in solution; q_e (mg/g) is the equilibrium adsorption capacity; Q_{max} (mg/g) is the maximum adsorption capacity per gram of sorbent; b (L/mg) is the Langmuir constant related to the energy of adsorption; K_f(mg/g) is Freundlich constants related to adsorption capacity and n is the Freundlich exponent, which indicates the adsorption intensity.

The pseudo-first-order and the pseudo-second-order models were applied to evaluate the adsorption kinetic data, which can be expressed as [27, 28]:

$$q_t = q_e(1 - e^{-k_1 t}) \tag{4}$$

$$q_t = \frac{k_2 q_e^2 t}{1 + k_2 q_e t} \tag{5}$$

Where q_e (mg/g) is the amount of adsorbed fluoride at equilibrium, and q_t (mg/g) is the amount of adsorbed fluoride at time t; k_1 (1/min) and k_2 (g/(mg min)) are the rate constant of the pseudo-first-order and pseudo-first-order model, respectively.

3. Results and Discussion

3.1 Structure characterization of the samples

The crystal structures of the samples were evaluated by XRD. Fig. 1 shows the XRD patterns of the samples by hydrothermal for different time. DCP-0.5 exhibited the main diffraction peaks of monoclinic brushite phases (JCPDF card no. 72-1240) mixed with a little triclinic monetite phases (JCPDF card no. 71-1759). With increasing the time to 1 h, the diffraction peaks assigned to brushite phases decreased (Fig. 1b), indicating that brushite was transformed into monetite by dehydration [24]. Further prolonging the reaction time, the diffraction peaks of hydroxylapatite (JCPDF card no. 70-0794) appeared in Fig. 1c, until DCP-10 were completely converted into the hydroxylapatite. This is due to the hydrolysis of monetite in alkali solution [29].

According to the above XRD results, it is obvious that the samples by hydrothermal for different time consist of the different compositions. Thus, it is necessary to assess the effect of hydrothermal time on the fluoride removal. As shown

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in Fig. 2, the samples displayed the distinct fluoride removal ability. The sample DCP-1 had the highest fluoride adsorption capacity, while the hydroxylapatite DCP-10 exhibited the worst adsorption ability. We could exclude the influence of BET area on the removal of fluoride, because BET area of DCP-1 (15.1 m^2/g) was inferior to those of DCP-3 (33.9 m^2/g) and DCP-10 (33.7 m^2/g), which was just slightly higher than that of DCP-0.5 (13.8 m^2/g). It was deduced that there must be other factor to control the adsorption behavior of DCP. By analyzing the composition of samples, the reason might be speculated that monetite, the main phase in DCP-1, could expose more adsorption sites towards fluoride, or it easily captured fluoride ions to form the news species. The hypothesis will be confirmed in the following investigation of adsorption mechanism. The sample DCP-1 was used for the further characterization and adsorption experiment, and DCP-1 was abbreviated to DCP unless otherwise specified in the following investigation.

The morphology and structure of DCP were investigated by SEM and TEM in Fig. 3. The SEM images (Fig. 3 a and b) showed that DCP samples consisted of irregular particles with a size of ca. 3 μm. The surface of DCP samples was relatively smooth, and some smaller particles appeared on this surface. Such large particles were presumably attributed to the agglomeration of small particles in the drying process. TEM images (Fig. 3 c and d) further demonstrated the existence of numerous smaller particles with a wide size distribution, revealing that smaller grains could aggregate into the large particles in the drying process.

3.2 Adsorption isotherm

The adsorption isotherm can evaluate the adsorption process, which describes the relationship between the adsorbate in the liquid phase and the adsorption sites on the adsorbent surface. The common Langmuir and Freundlich isotherms are used in the present study [26, 30]. The results of fitting curves are shown in Fig. 4a, and the corresponding calculated parameters are listed in Table 1. It was observed that the adsorption capacity of fluoride increased with increasing the fluoride concentration.

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The correlation coefficient (R^2) values proved that the Langmuir isotherm model is more suitable than the Freundlich isotherm model for describing the adsorption process. This suggested that the fluoride should be adsorbed on the homogeneous surface of absorbent in the form of monolayer. In addition, the positive Langmuir constant *b* and the high Freundlich constant n (>1) implied the good affinity of DCP adsorbent towards fluoride. The theoretical maximum adsorption capacity of fluoride from the Langmuir isotherm model is 66.72 mg/g, which is higher than that of other calcium phosphate reported in Table 2.

3.3 Effect of initial pH

The solution pH has a vital influence on the surface charge of the adsorbent and the dissociation of surface functional groups. The effect of initial pH on fluoride adsorption is presented in Fig. 4b. It could be seen that the fluoride adsorption capacity reached the maximum value 40.5 mg/g at pH 3. At low pH 2.1, the adsorption capacity dramatically decreased to 6.4 mg/g, and kept constant 26 mg/g in the pH rang of 6-10. It is generally accepted that the electrostatic interaction and ion-exchange dominated the adsorption ability of calcium-based adsorbent toward fluoride [27, 36]. The protonation enhanced the electrostatic interaction between the positive charged surface and the negative charged fluoride at low pH, thereby improving the adsorption ability of DCP. However, DCP took the risk of dissolving at lower pH, resulting in the reduction of adsorption sites as well as the abatement of electrostatic interaction due to the formation of hydrofluoric acid (HF). While with the increase of pH, the degree of surface protonation would decrease gradually, and the negative charge surface would be not beneficial to the electrostatic interaction with the negative anionic fluoride ions. In addition, the large amounts of OH^- ions might compete with fluoride ions for the adsorption sites [37]. These reasons will give rise to the reduction of fluoride adsorption capacity. It is worth noting that pH value had the neglected influence on the adsorption capacity in the pH range of 6-10. This

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means that there must be other factor to control the adsorption behavior besides electronic interaction and ions exchange, such as chemical reaction.

3.4 Adsorption kinetics

The adsorption rate is a crucial criterion to evaluate the adsorption efficiency. The effect of contact time on fluoride adsorption was shown in Fig. 5a. It was observed that the adsorption capacity quickly increased in the initial 60 min, subsequently increased at a slower rate and reached the adsorption equilibrium in 240 min. The kinetic data was further fitted by the pseudo-first-order and the pseudo-second-order models. As shown in Fig. 5b, the plots can be fitted well by pseudo-second-order model with the high R^2 value 0.962 (pseudo-first-order value: 0.254), implying that the rate limiting step may be chemisorption involving valence forces by sharing or electron exchange between adsorbent and adsorbate [38, 39]. The adsorption capacity q_e (30.96 mg/g) obtained from the kinetic curve coincided with the experimental value (28.92 mg/g).

3.5 Adsorption thermodynamics and effect of co-existing ions

The thermodynamic parameters, such as standard Gibbs free energy (ΔG^0), standard enthalpy change (ΔH^0), and standard entropy change (ΔS^0) were calculated to evaluate the thermodynamic feasibility and the nature of adsorption process according to the following equations [40]:

$$K_0 = \frac{q_e}{C_e} \tag{6}$$

$$\Delta G^0 = -RT \ln K_0 \tag{7}$$

$$\ln K_0 = \frac{\Delta S^0}{R} - \frac{\Delta H^0}{RT} \tag{8}$$

where K_0 is the distribution coefficient (mL/g); R is the universal gas constant; T is the absolute temperature (K). The values of ΔH^0 and ΔS^0 were obtained from the slope and intercept of $\ln K_0$ against $1/T$.

The thermodynamics plots are displayed in Fig. 6a, and the parameters are listed in Table 3. It was seen that ΔG^0 at all evaluated temperature exhibited the negative

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values, demonstrating the feasible and spontaneous adsorption process, and the degree of spontaneity increased with the increase of temperature. The adsorption process was endothermic in nature, as confirmed by the positive ΔH^0 value (4.48 kJ/mol). It was deduced that chemical adsorption should be involved in the whole process, implying the occurrence of chemical reaction. Moreover, the positive ΔS^0 (36.37 J/mol·K) reflected the affinity of adsorbent towards fluoride.

In natural water environment, several common anions, such as Cl^-, NO_3^-, SO_4^{2-}, PO_4^{3-} and CO_3^{2-}, had a significant influence on the fluoride adsorption, which can compete with fluoride for adsorption active sites and decrease the adsorption efficiency. The effect of co-existing anions was performed in the presence of 100 mg/L Cl^-, 20 mg/L NO_3^-, 80 mg/L SO_4^{2-}, 50 mg/L PO_4^{3-} and 50 mg/L CO_3^{2-} with a 36 mg/L initial fluoride concentration. As shown in Fig. 6b, the co-existing anions had a negligible influence on the fluoride adsorption, and the fluoride adsorption capacities decreased from 30.9 to ca. 28 mg/L after the addition of co-existing anion. The results indicated that the DCP adsorbent had the strong affinity towards fluoride ions compared with other common anions.

3.6 Adsorption mechanism

XRD can reflect the crystal structure change of the as-prepared samples before and after fluoride adsorption. As shown in Fig. 7a, the DCP sample underwent the tremendous crystal structure change after fluoride adsorption, which is totally different from the pristine DCP sample. Fluoride adsorption onto the DCP sample resulted in the conversion of adsorbent from monetite to fluorapatite (JCPDF card no. 15-0876), implying that the chemical reaction and/or dissolution – precipitation mechanism took place between DCP sample and fluoride ions [41]. It is also deduced that the fluoride ions can easily migrate to the surface of DCP sample from the solution, further reacting with the DCP to form the news species, which is consistent with the as-mentioned hypothesis. EDX of DCP and DCP loaded fluoride are displayed in Fig. 7b and c. It was clear that the fluoride ions appeared on the surface of DCP after fluoride adsorption, which confirmed the occurrence of fluoride

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migration. The migration was presumably accompanied by the little exchange between fluoride ions and hydroxide ions, otherwise, fluoride ions would hardly be close to the inside of DCP.

The FTIR spectra of DCP and DCP loaded fluoride are shown in Fig. 8. For the DCP spectrum, the bands at 527 and 580 cm^{-1} were attributed to O-P-O(H) bending mode, while the band at 899 cm^{-1} was assigned to O-P-O(H) stretching vibration mode [42]. The P-O stretching mode appeared at 1004, 1065 and 1131 cm^{-1}. The band at 1403 cm^{-1} was related to the P-O-H in-plane bending. The above mentioned bands corresponded to the characteristic bands of PO_4^{3-}, confirming the monetite structure. After fluoride loaded onto DCP, the bands at 473, 566, 605, 965 1036 and 1097 cm^{-1} belonged to the stretching vibrations of tetrahedral PO_4^{3-}. The locations of these bands are in good agreement with the fluorapatite of the literature reports [43, 44], directly demonstrating the formation of new species fluorapatite after fluoride loaded onto DCP.

XPS characterization is used to investigate the fluoride adsorption mechanism onto DCP. As displayed in Fig. 9a, the wide scan XPS spectra showed that the peaks at about 133, 284, 347 and 531 eV were attributed to P 1s, C 1s, Ca 2p and O 1s, respectively. After fluoride loaded onto DCP, a new peak at 648.1 eV appeared, which corresponded to F 1s, indicating the migration of fluoride ions onto the DCP from the solution. Fig. 9b shows the Ca 2p spectrum of DCP and DCP loaded fluoride. It is obvious that the binding energy of Ca 2p had a about 0.2 eV shift after fluoride loaded onto DCP, revealing the strong interaction between fluoride and Ca atoms.

4. Conclusions

Dicalcium phosphate adsorbent was synthesized by a facile method, and it could effectively be used to remove the fluoride ions from aqueous solution. XRD result indicated that monetite was the dominant phase in the as-prepared dicalcium phosphate. The adsorbent showed the relatively high adsorption capacity in the wide pH range of 3-10. The adsorption of fluoride onto DCP followed the Langmuir

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isotherm model with a maximum adsorption capacity of 66.72 mg/g, which is higher than other calcium phosphate-based materials. The adsorption kinetic data could be described by pseudo-second-order kinetic model. The co-existing anions had a negligible influence on the fluoride adsorption. The adsorption mechanism was investigated by XRD, FTIR and XPS, and the chemical reaction and/or dissolution – precipitation mechanism, dominated the whole adsorption process, in conjunction with electronic interaction and ions exchange, cooperatively enhanced the fluoride removal.

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Reference:

1. Bhatnagar A, Kumar E, Sillanpaa M. Fluoride removal from water by adsorption-A review. Chem Eng J. 2011;171:811-40.
2. Chi YL, Chen YT, Hu CL, Wang YS, Liu C. Preparation of Mg-Al-Ce triple-metal composites for fluoride removal from aqueous solutions. J Mol Liq. 2017;242:416-22.
3. Xiang W, Zhang GK, Zhang YL, Tang DD, Wang JT. Synthesis and characterization of cotton-like Ca-Al-La composite as an adsorbent for fluoride removal. Chem Eng J. 2014;250:423-30.
4. Sternitzke V, Kaegi R, Audinot JN, Lewin E, Hering JG, Johnson CA. Uptake of fluoride from aqueous solution on nano-sized hydroxyapatite:examination of a fluoridated surface layer. Environ Sci Technol. 2012;46:802-9.
5. Reardon EJ, Wang YX. A limestone reactor for fluoride removal from wastewaters. Environ Sci Technol. 2000;34:3247-53.
6. Lahnid S, Tahaikt M, Elaroui K, Idrissi I, Hafsi M, Laaziz I, Amor Z, Tiyal F, Elmidaoui A. Economic evaluation of fluoride removal by electrodialysis. Desalination. 2008;230:213-9.
7. Ghosh D, Medhi CR, Purkait MK. Treatment of fluoride containing drinking water by electrocoagulation using monopolar and bipolar electrode connections. Chemosphere. 2008;73:1393-400.
8. Meenakshi S, Viswanathan N. Identification of selective ion-exchange resin for fluoride sorption. J Colloid Interface Sci. 2007;308:438-50.
9. Ndiayea PI, Moulin P, Dominguez L, Millet JC, Charbit F. Removal of fluoride from electronic industrial effluent by RO membrane separation. Desalination. 2005;173:25-32.
10. Agarwal M, Rai K, Shrivastav R, Dass S. Deflouridation of water using amended clay. J Clean Prod. 2003;11:439-44.

ACCEPTED MANUSCRIPT

11. Xu Y, Chai X. Characterization of coal gasification slag-based activated carbon and its potential application in lead removal. Environ Technol. 2018;39:382-91.
12. Wu LY, Zhang GK, Tang DD. A novel high efficient Mg-Ce-La adsorbent for fluoride removal: kinetics, thermodynamics and reusability. Desalination and Water Treatment. 2016;57:23844-55.
13. Kang DJ, Yu XL, Ge MF. Morphology-dependent properties and adsorption performance of CeO2 for fluoride removal. Chem Eng J. 2017;330:36-43.
14. Kanno CM, Sanders RL, Flynn SM, Lessard G, Myneni SCB. Novel Apatite-Based Sorbent for Defluoridation: Synthesis and Sorption Characteristics of Nano-micro-crystalline Hydroxyapatite-Coated-Limestone. Environ Sci Technol. 2014;48:5798-807.
15. Yu XL, Kang DJ, Hu YY, Tong SR, Ge MF, Cao CY, Song WG. One-pot synthesis of porous magnetic cellulose beads for the removal of metal ions. RSC Adv. 2014;4:31362-9.
16. Tang DD, Zhang GK. Efficient removal of fluoride by hierarchical Ce-Fe bimetal oxides adsorbent: Thermodynamics, kinetics and mechanism. Chem Eng J. 2016;283:721-9.
17. Kang DJ, Tong SR, Yu XL, Ge MF. Template-free synthesis of 3D hierarchical amorphous aluminum oxide microspheres with broccoli-like structure and their application in fluoride removal. RSC Adv. 2015;5:19159-65.
18. Wu X, Zhang Y, Dou X, Yang M. Fluoride removal performance of a novel Fe-Al-Ce trimetal oxide adsorbent. Chemosphere. 2007;69:1758-64.
19. Mirkovic MM, Pasti TDL, Dosen AM, Cebela MZ, Rosic AA, Matovic BZ, Babic BM. Adsorption of malathion on mesoporous monetite obtained by mechanochemical treatment of brushite. RSC Adv. 2016;6:12219-25.
20. Zhang DY, Luo HM, Zheng LW, Wang KJ, Li HX, Wang Y, Feng HX. Utilization of waste phosphogypsum to prepare hydroxyapatite nanoparticles and its application towards removal of fluoride from aqueous solution. J Hazard Mater. 2012;241:418-26.
21. Mourabet M, El Boujaady H, El Rhilassi A, Ramdane H, Bennani-Ziatni M, El Hamri R, Taitai A. Defluoridation of water using Brushite: Equilibrium, kinetic and thermodynamic studies. Desalination. 2011;278:1-9.
22. Jayarathne A, Weerasooriya R, Chandrajith R. A rapid method for the removal of fluoride in contaminated groundwater using natural crystalline apatite: a laboratory and field study. Environmental Earth Sciences. 2015;73:8369-77.
23. Sahin E, Ciftcioglu M. Monetite promoting effect of NaCl on brushite cement setting kinetics. Journal of Materials Chemistry B. 2013;1:2943-50.
24. Chen S, Grandfield K, Yu S, Engqvist H, Xia W. Synthesis of calcium phosphate crystals with thin nacreous structure. Crystengcomm. 2016;18:1064-9.
25. Shen CS, Wu LX, Chen YH, Li S, Rashid S, Gao Y, Liu JS. Efficient removal of fluoride from drinking water using well-dispersed monetite bundles inlaid in chitosan beads. Chem Eng J. 2016;303:391-400.
26. Yu XL, Tong SR, Ge MF, Zuo JC, Cao CY, Song WG. One-step synthesis of magnetic composites of cellulose@iron oxide nanoparticles for arsenic removal. J Mater Chem A. 2013;1:959-65.
27. Yu XL, Tong SR, Ge MF, Zuo JC. Removal of fluoride from drinking water by cellulose@hydroxyapatite nanocomposites. Carbohydr Polym. 2013;92:269-75.
28. Singh SK, Townsend TG, Mazyck D, Boyer TH. Equilibrium and intra-particle diffusion of stabilized landfill leachate onto micro- and meso-porous activated carbon. Water Res. 2012;46:491-9.

ACCEPTED MANUSCRIPT

29. Yu YD, Zhu YJ, Qi C, Wu J. Hydroxyapatite nanorod-assembled hierarchical microflowers: rapid synthesis via microwave hydrothermal transformation of CaHPO4 and their application in protein/drug delivery. Ceram Int. 2017;43:6511-8.
30. Wang M, Yu XL, Yang CL, Yang XQ, Lin MY, Guan LT, Ge MF. Removal of fluoride from aqueous solution by Mg-Al-Zr triple-metal composite. Chem Eng J. 2017;322:246-53.
31. Ruan ZY, Tian YX, Ruan JF, Cui GJ, Iqbal KW, Iqbal A, Ye HR, Yang ZZ, Yan SQ. Synthesis of hydroxyapatite/multi-walled carbon nanotubes for the removal of fluoride ions from solution. Appl Surf Sci. 2017;412:578-90.
32. Carslaw DC, Rhys-Tyler G. New insights from comprehensive on-road measurements of NOx, NO2 and NH3 from vehicle emission remote sensing in London, UK. Atmos Environ. 2013;81:339-47.
33. Nie YL, Hu C, Kong CP. Enhanced fluoride adsorption using Al (III) modified calcium hydroxyapatite. J Hazard Mater. 2012;233:194-9.
34. Chen L, Zhang KS, He JY, Xu WH, Huang XJ, Liu JH. Enhanced fluoride removal from water by sulfate-doped hydroxyapatite hierarchical hollow microspheres. Chem Eng J. 2016;285:616-24.
35. Pandi K, Viswanathan N. In situ precipitation of nano-hydroxyapatite in gelatin polymatrix towards specific fluoride sorption. Int J Biol Macromol. 2015;74:351-9.
36. Chen YH, Shen CS, Rashid S, Li S, Ali BA, Liu JS. Biopolymer-induced morphology control of brushite for enhanced defluorination of drinking water. J Colloid Interface Sci. 2017;491:207-15.
37. Zhang L, Liu X, Huang X, Wang W, Sun P, Li Y. Adsorption of Pb2+ from aqueous solutions using Fe-Mn binary oxides-loaded biochar : Kinetics, isotherm and thermodynamic studies. Environ Technol. 2018:1-24.
38. Kang D, Yu X, Ge M, Xiao F, Xu H. Novel Al-doped carbon nanotubes with adsorption and coagulation promotion for organic pollutant removal. Journal of Environmental Sciences. 2017;54:1-12.
39. Toor M, Jin B. Adsorption characteristics, isotherm, kinetics, and diffusion of modified natural bentonite for removing diazo dye. Chem Eng J. 2012;187:79-88.
40. Kang D, Yu X, Ge M, Lin M, Yang X, Jing Y. Insights into adsorption mechanism for fluoride on cactus-like amorphous alumina oxide microspheres. Chem Eng J. 2018;345:252-9.
41. Hokkanen S, Bhatnagar A, Sillanpaa M. A review on modification methods to cellulose-based adsorbents to improve adsorption capacity. Water Res. 2016;91:156-73.
42. Tortet L, Gavarri JR, Nihoul G, Dianoux AJ. Study of protonic mobility in CaHPO4 center dot 2H(2)O (brushite) and CaHPO4 (monetite) by infrared spectroscopy and neutron scattering. J Solid State Chem. 1997;132:6-16.
43. Antonakos A, Liarokapis E, Leventouri T. Micro-Raman and FTIR studies of synthetic and natural apatites. Biomaterials. 2007;28:3043-54.
44. Karimi M, Ramsheh MR, Ahmadi SM, Madani MR, Shamsi M, Reshadi R, Lotfi F. Reline-assisted green and facile synthesis of fluorapatite nanoparticles. Mater Sci Eng C-Mater Biol Appl. 2017;77:121-8.

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Table 1 Langmuir and Freundlich parameters of fluoride adsorption on DCP adsorbent

Langmuir parameters			Freundlich parameters		
Q	b (L/mg)	R^2	K_f (mg/g)	n	R^2
66.72	0.093	0.976	17.88	3.73	0.963

Table 2 Comparison of fluoride adsorption capacity with other calcium phosphate

Adsorbents	Q (mg/g)	Reference
Monetite particles inlaid chitosan	50.01	[25]
Hydroxyapatite/multi-walled carbon nanotubes	30.22	[31]
Cellulose@hydroxyapatite nanocomposites	4.2	[32]
Hydroxyapatite nanoparticles	19.7	[20]
Brushite	6.59	[21]
Al modified calcium	32.57	[33]
Sulfate-doped hydroxyapatite microspheres	28.3	[34]
Hydroxyapatite-coated-limestone	8.8	[14]
n-HAp@Gel composite	4.16	[35]
Dicalcium phosphate	66.72	This study

Table 3 Thermodynamic parameters for fluoride adsorption on adsorbent

T (K)	ΔG^0 (kJ/mol)	ΔS^0 (J/mol K)	ΔH^0 (kJ/mol)
298	-3.38	36.37	4.48
308	-3.62		

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313	-3.77
318	-3.91

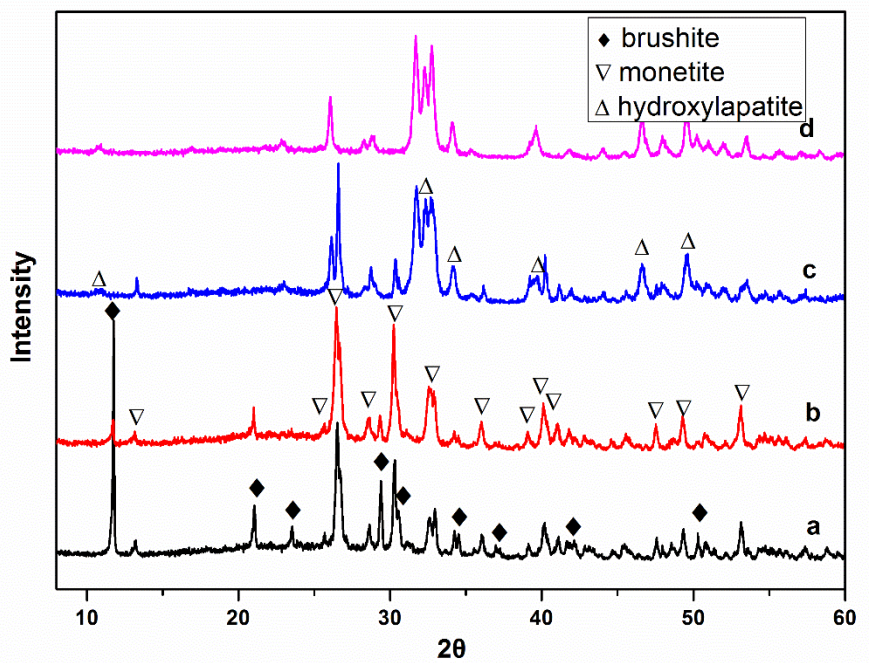

Fig. 1 XRD patterns of the samples by hydrothermal for 0.5 h (a), 1 h (b), 3 h (c) and 10 h (d).

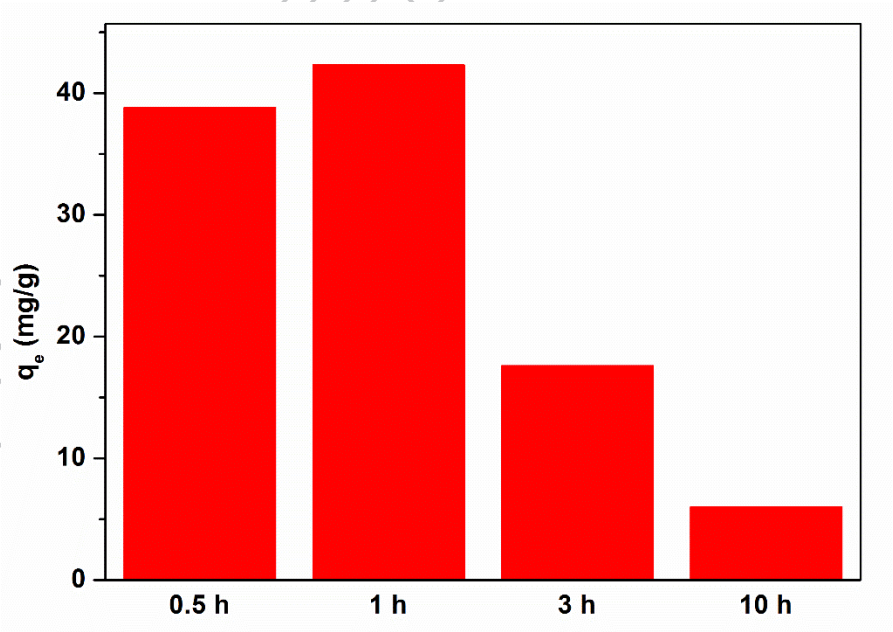

Fig. 2 Effect of hydrothermal time on the fluoride removal.

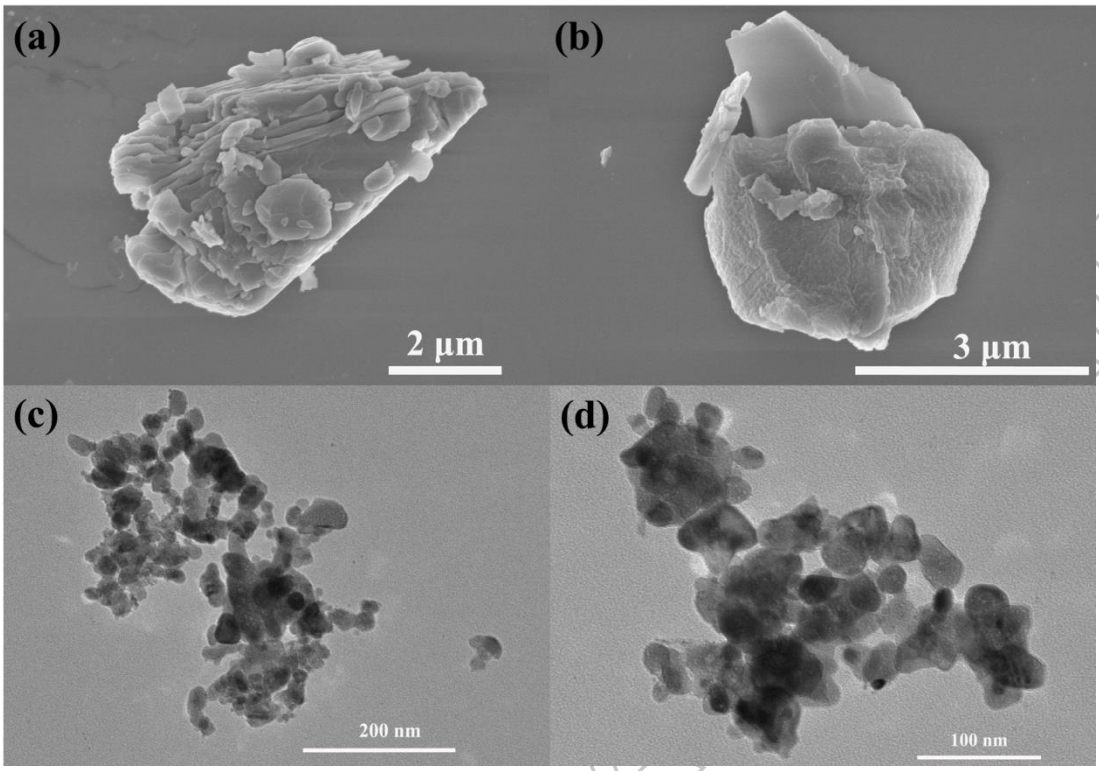

Fig. 3 SEM (a and b) and TEM (c and d) images of DCP

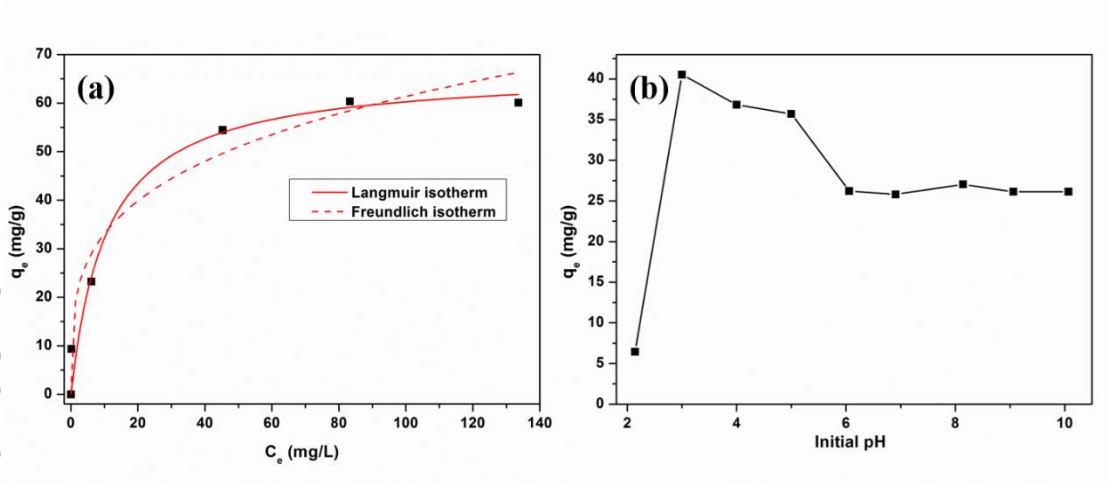

Fig. 4 Langmuir and Freundlich isotherm for fluoride adsorption (a), and efect of initial pH on fluoride adsorption (b)

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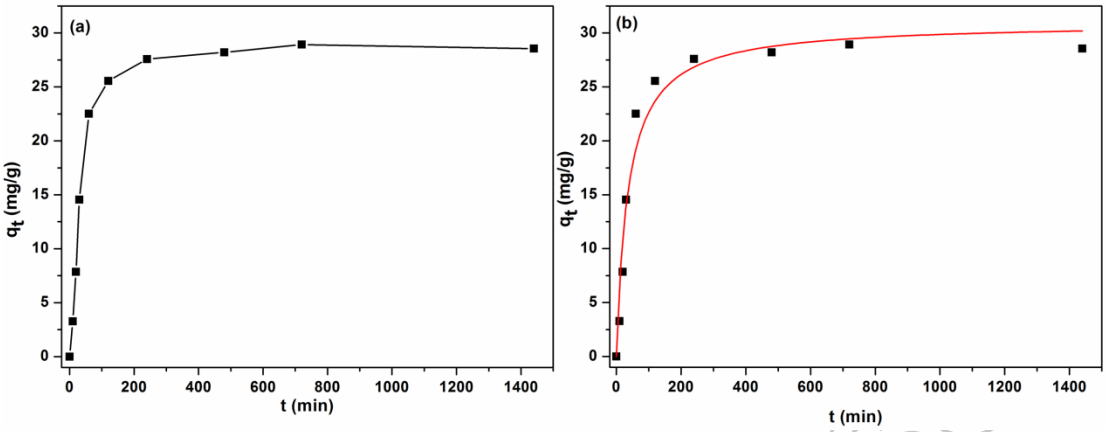

Fig. 5 Adsorption kinetics of fluoride onto DCP: (a) effect of contact time; (b) fitting curve of pseudo-second-order model

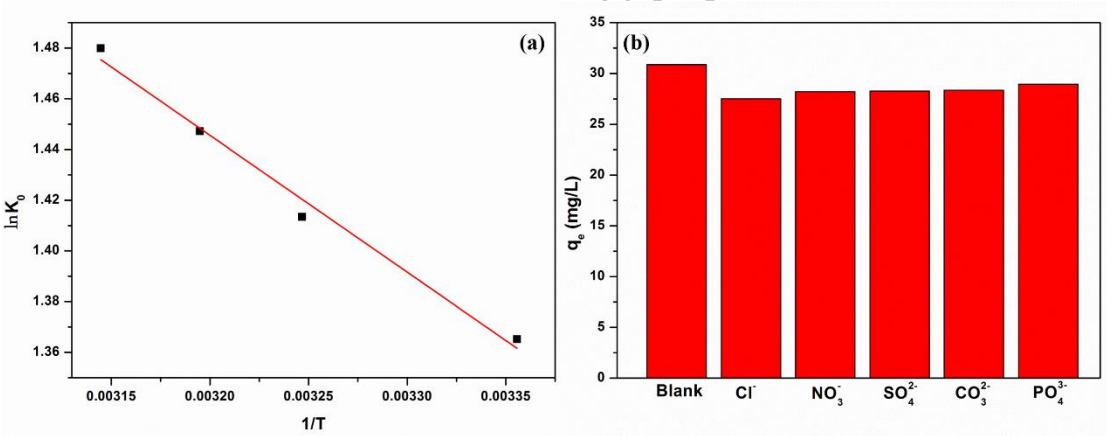

Fig. 6 Thermodynamics plots (a) and effect of co-existing ions (b) for fluoride adsorption onto the adsorbent

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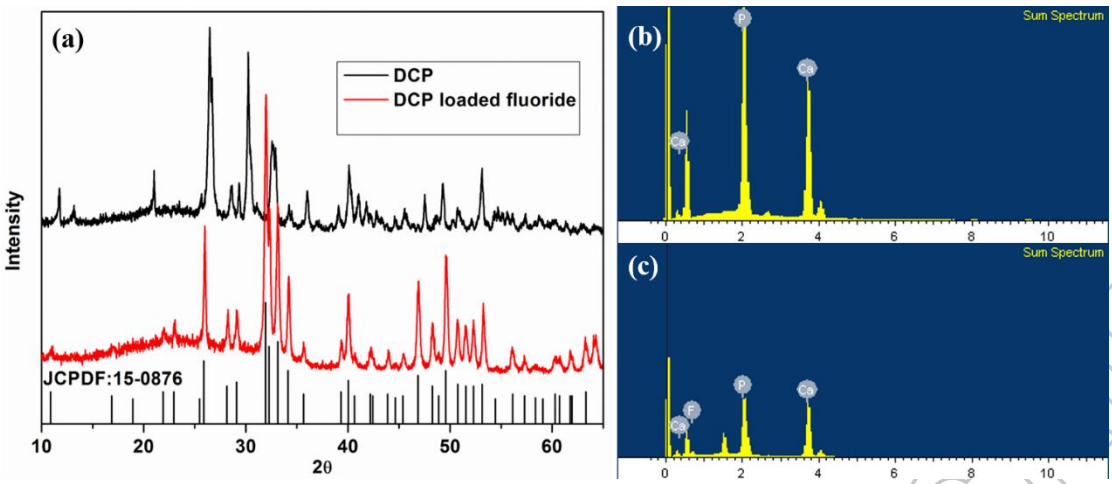

Fig. 7 XRD patterns (a) and EDX (b and c) of DCP and DCP loaded fluoride

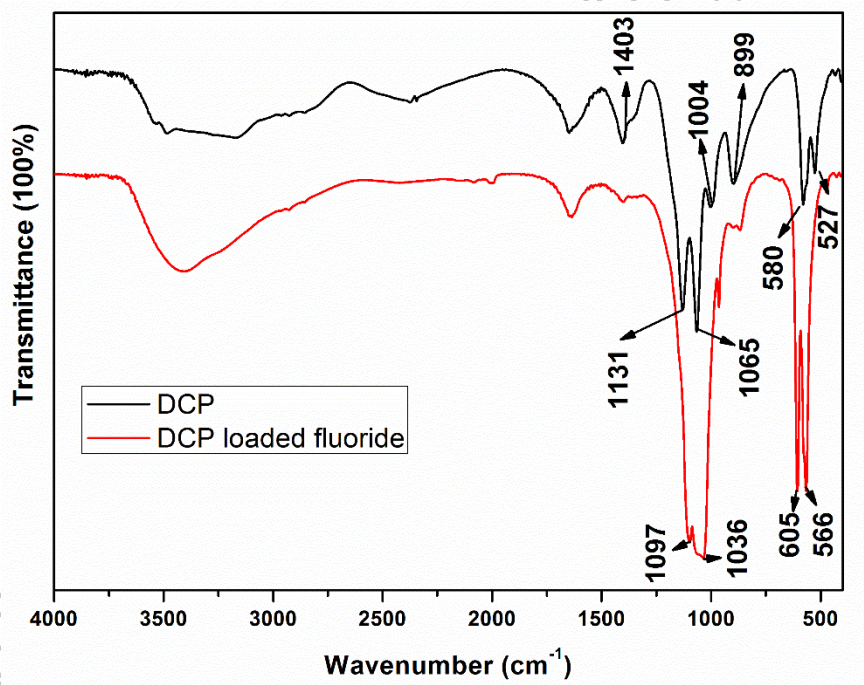

Fig. 8 FTIR spectra of DCP and DCP loaded fluoride

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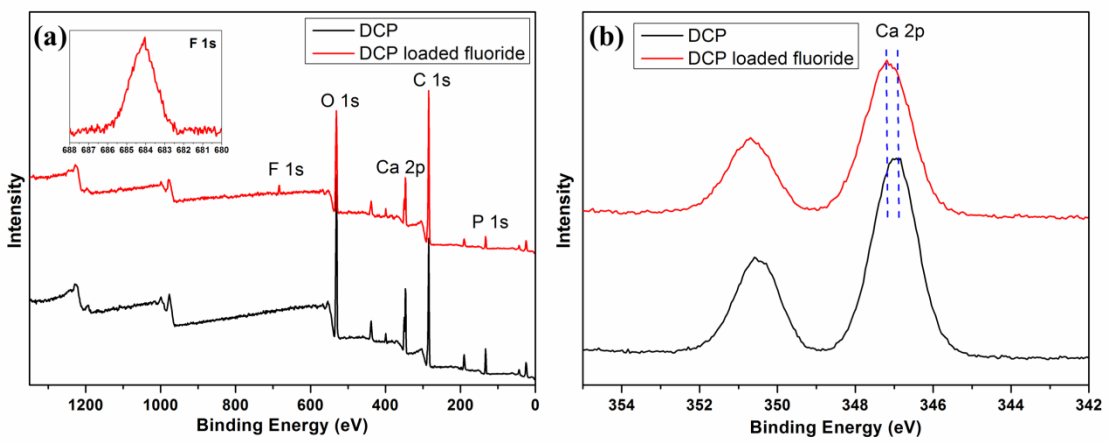

Fig. 9 XPS curves of DCP and DCP loaded fluoride: (a) The wide scan spectra and inset is enlarged XPS spectrum of fluoride; (b) Ca 2p spectra.

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Figure captions

Fig. 1 XRD patterns of the samples by hydrothermal for 0.5 h (a), 1 h (b), 3 h (c) and 10 h (d).

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